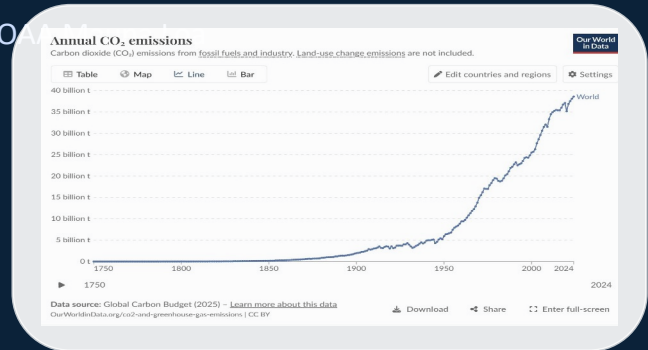


Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024 emissions; atmospheric CO₂ to Feb 2026

Sources: Global Carbon Budget 2025 (via Our World in Data); NASA Vital Signs; NOAA



Atmospheric CO₂ Milestone

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

Latest Mauna Loa (NOAA) reading

429 ppm

February 2026. ~50% above the pre-industrial baseline (~280 ppm).

Context

1.5×

At least 1.5× greater than the natural increase observed at the end of the last ice age (20,000 years ago).

Acceleration since 1958 (Mauna Loa):



Primary driver: burning coal, oil, and natural gas (confirmed by ground + satellite measurements).

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

Line chart (description): annual global CO₂ emissions from fossil fuels + industry, 1750–2024.

1950

~6 Bt

1990

>20 Bt

Today

35+ Bt

Note: “figure_2.jpg” referenced in outline is not present in provided assets; slide uses a

described chart instead.
Source: Global Carbon Budget 2025 (via Our World in Data).

Growth slowed, but not yet

Country-Level Totals

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

- China: ~12.5 Bt (~27%) — slight increase
- United States: ~5.0 Bt (~14%) — declining (peaked near 6 Bt around 2005)
- India: ~3.0 Bt (~8%) — rising
- EU-27: ~2.7 Bt (~7%) — declining

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

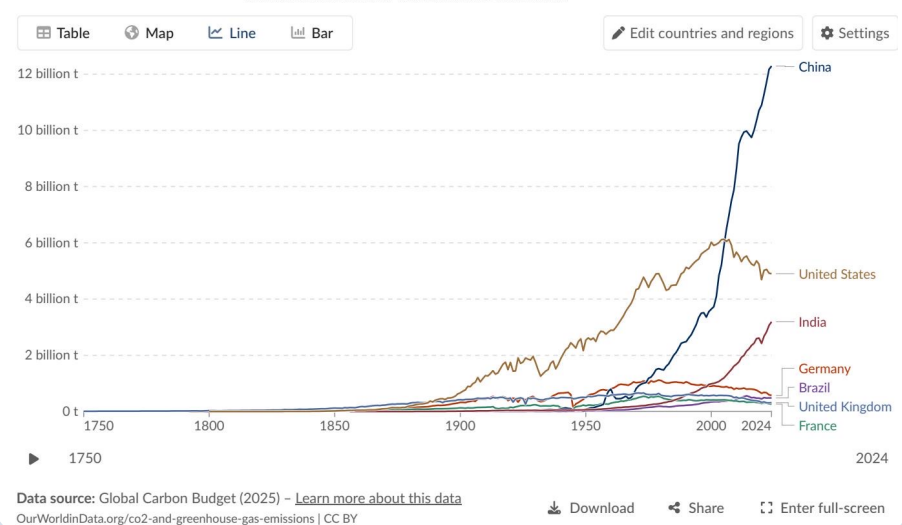


Figure: annual CO₂ emissions by country (reference: figure 3.png).

Regional Share Shift

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900 (share of global emissions)

Europe + US > 90%

Others (small share)

2024 (share of global emissions)

Asia ~50%

US + Europe < 33%

Africa 3–4%

S. America 3–4%

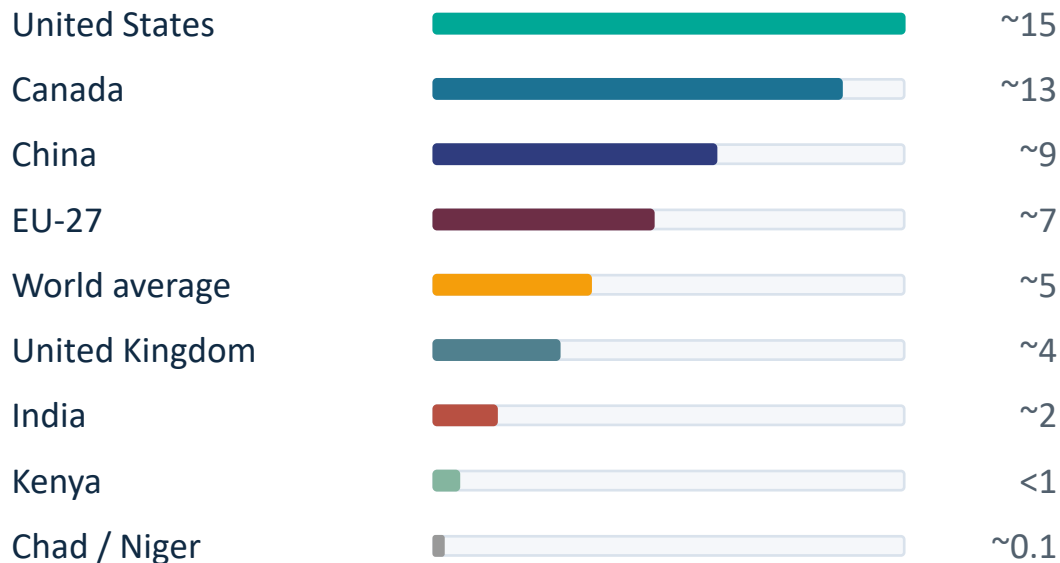
Visual to use: stacked area (1900→2024) or side-by-side share blocks (shown). Key message: emissions' center shifted from Europe+US

Source: Global Carbon Budget 2025 (via Our World in Data).

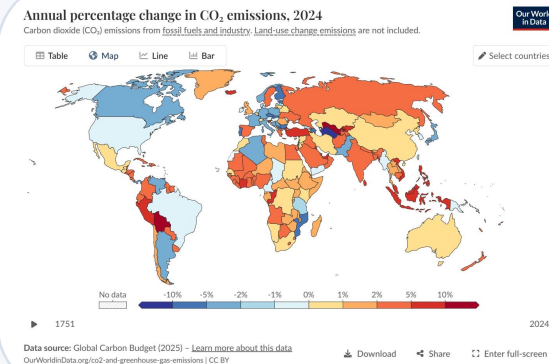
Per Capita Inequality

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

Annual CO₂ per person (tonnes)



Source: Global Carbon Budget 2025 (via Our World in Data).



Reference: per-capita line chart (figure_1.jpg)

~150× gap

Average American/Australian emits in
<2 days what the average person in

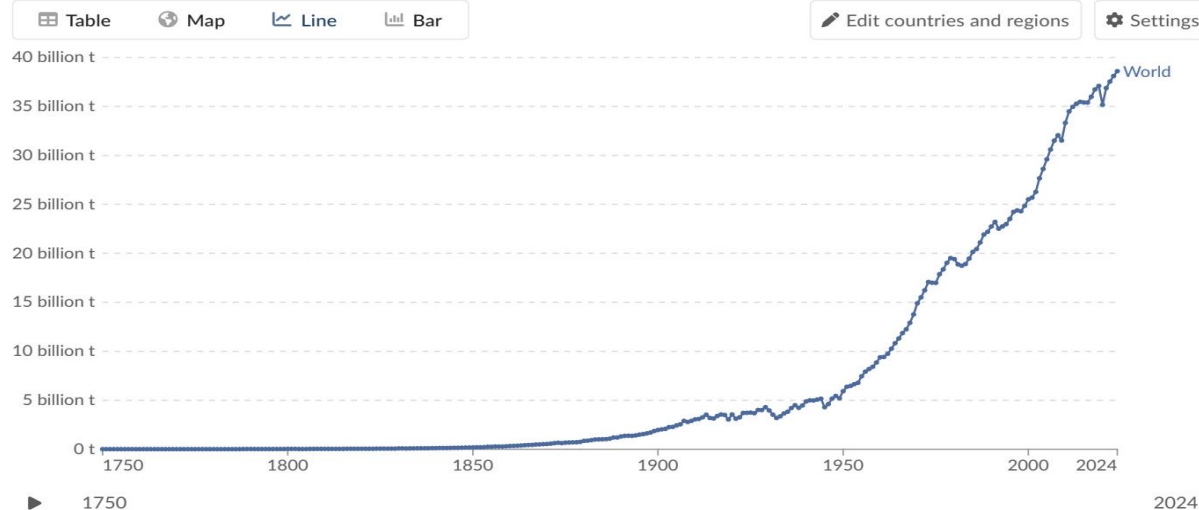
Chad/Niger emits in a year.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data



Data source: Global Carbon Budget (2025) - [Learn more about this data](#)



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How to read (map legend)

- US + much of Europe: declines ($\approx -1\%$ to -5%)
- Russia: increases
- Parts of SE Asia: increases
- Several African nations: +2% to +10% or more
- Some South American countries: >10% increases (others decline)

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Key point: similar living standards can produce very different per-capita CO₂ outcomes — driven by energy sources.

- France + UK: higher nuclear and renewables → lower per-capita CO₂
- Germany + Netherlands: higher fossil share → higher per-capita CO₂
- Implication for negotiations: policy and technology pathways matter, not prosperity alone.

Group	Dominant sources (qualitative)
France / UK	Higher nuclear + renewables
Germany / Netherlands	Higher fossil fuel share

Data constraint note: the brief provides France/UK per-capita CO₂ values and a qualitative comparison to Germany/Netherlands (higher for Germany/Netherlands). The brief does not include numeric per-capita values for Germany/Netherlands.

Key Takeaways

- Global CO₂ emissions exceed 35 Bt/yr and have not yet peaked (growth has slowed).
- China emits ~12.5 Bt (~27%) — more than double the United States (~5 Bt).
- Regional responsibility has shifted: Asia ~50%; US + Europe combined < one-third.
- Per-capita inequality remains extreme (~150× between ~15 t and ~0.1 t per person).
- Energy mix and policy choices can decouple prosperity from emissions intensity.